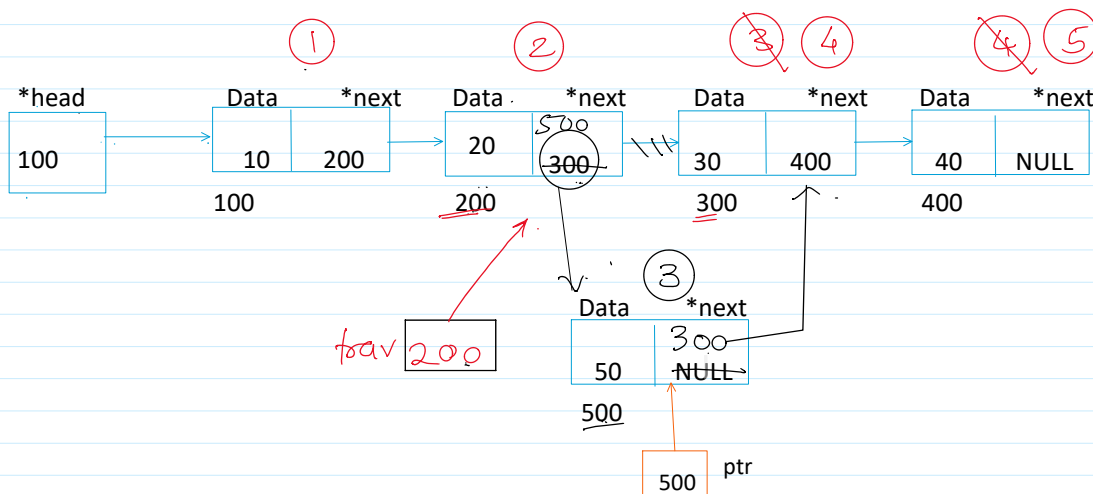


pos = 4 X

If List is empty

If (head == NULL)
 {
 If (pos == 1)
 addnode_at_first_pos(value);
 Else
 printf("Cannot add node at this pos");
 }
 If pos == 1
 addnode_at_1st_pos();
 If pos == count+1
 8 == 7+1
 addnode_at_last_pos();
 If pos < 1 or pos > count+1
 printf("Cannot add node at this pos");

- ① Create a node
- ② Assign the value
- ③ Attach.

pos → 3

1) Take a trav pointer starting from the first node.
Struct node *trav = head;

2) Traverse till pos-1 node.
For(int i=1; i<pos-1; i++)
 Trav = trav->next;

3) connect the new node to the current pos node(3rd node) of the linked list.
Ptr->next = trav->next;

4) Connect the pos-1 node to the new node.
Trav->next = ptr;

pos
 500 → next = 300;
 ptr → next = pos-1 → next;
 trav

ptr → next = trav → next;
 2nd →

200 → next = 500;
 trav → next = ptr;

Best Case : pos == 1: $O(1)$

Worst Case : pos == count+1 : $O(n)$

Avg Case : if pos no of node to be added is somewhere in between : $O(n)$